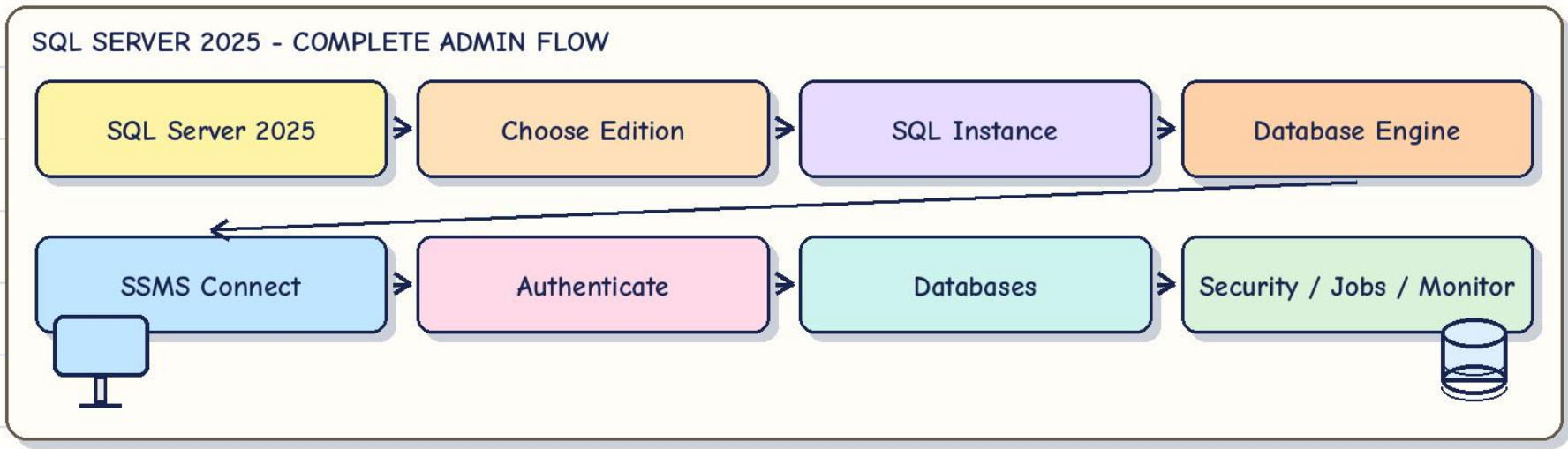
REVISION CHEAT SHEET:

SQL SERVER EDITIONS & ADMIN TOOLS

Editions • SSMS • Security • Agent • Monitoring

SQL Server Editions, SSMS, Connections, Security, SQL Server Agent, and Activity Monitor



Before learning each topic separately, understand the overall picture:

```text
SQL Server 2025
  ↓
Choose an Edition
  ↓
Install SQL Server Instance
  ↓
Database Engine runs
  ↓
Use SSMS to connect
  ↓
Authenticate yourself
  ↓
Access databases
  ↓
Manage databases, security, jobs, performance
```

For example:

```text
SQL Server 2025 Developer Edition
      ↓
   DEV-PC\SQLDEV
      ↓
   Database Engine
      ↓
    EmployeeDb
      ↑
```

TEXT (CONT.)

```
    SSMS
```

SQL Server Version vs SQL Server Edition

These two terms are different.

A version tells you which generation/release of SQL Server you are using.

Examples:

TEXT

```
SQL Server 2019
SQL Server 2022
SQL Server 2025
```

An edition tells you which feature/licensing level of that version you are using.

Examples:

TEXT

```
Enterprise
Standard
Enterprise Developer
Standard Developer
Express
Evaluation
```

So you can have:

TEXT

```
SQL Server 2025 Enterprise

SQL Server 2025 Standard

SQL Server 2025 Enterprise Developer

SQL Server 2025 Express
```

Simple memory:

TEXT

```
Version
=
Which generation?

Edition
=
Which package/tier?
```

SQL Server 2025 (17.x) is the current major SQL Server generation, and Microsoft provides Enterprise,

Standard, Enterprise Developer, Standard Developer, Evaluation, and Express editions.

SQL Server Editions

Different organizations have different requirements.

A small learning application does not need the same SQL Server capabilities as a huge enterprise system processing millions of business transactions.

That is why Microsoft provides different editions.

Think:

```text
Same SQL Server product
    ↓
Different editions
    ↓
Different features
Different limits
Different licensing
Different intended uses
```

Enterprise Edition

Enterprise Edition is the highest-end production edition.

It is designed for demanding enterprise systems requiring high levels of:

```text
Performance
Scalability
Availability
Security
Large workloads
Advanced SQL Server features
```

Typical use:

```text
Large banking system
Large healthcare system
Large enterprise platform
Mission-critical business system
```

Microsoft positions Enterprise as its premier SQL Server edition for demanding mission-critical workloads.

When to Use Enterprise

Use it when an organization genuinely requires Enterprise-only capabilities, very large scale, or advanced production features.

When Not to Use It

Do not choose Enterprise simply because:

```text
"Enterprise must always be better."
```

If Standard provides everything the application requires, Enterprise may introduce unnecessary licensing cost.

Standard Edition

Standard Edition is a production edition intended for organizations that need strong relational database capabilities without requiring every Enterprise feature.

Think:

```text
Enterprise
=
Highest feature level

Standard
=
Strong general-purpose production edition
```

SQL Server 2025 Standard supports up to the lesser of 4 sockets or 32 cores for the Database Engine and up to 256 GB for the buffer pool per instance.

Typical uses include:

```text
Business applications
Department systems
Medium-sized enterprise systems
.NET Web APIs
Internal company applications
```

SQL Server Developer Edition

This edition is particularly important for you as a learner and developer.

SQL Server 2025 provides:

```text
Enterprise Developer
Standard Developer
```

Enterprise Developer Edition

Enterprise Developer provides the functionality of Enterprise Edition but is licensed only for:

```text
Development
Testing
Learning
```

It is not licensed for production use.

That makes it excellent for learning SQL Server.

You can practice advanced features without buying an Enterprise production license.

Think:

```text
Enterprise Developer
=
Enterprise features

BUT

Development/Test only
```

Standard Developer Edition

SQL Server 2025 also provides Standard Developer Edition.

It contains the functionality of Standard Edition but, like Enterprise Developer, is licensed only for:

```text
Development
Testing
```

not production.

This is useful when developers want their development environment to behave more like a production environment using Standard Edition.

Which Developer Edition Should You Use for Learning?

For general SQL Server learning:

> **Enterprise Developer Edition is normally the best learning choice.**

Why?

Because you can experiment with the broad SQL Server feature set without production licensing costs.

For your learning machine:

```text
SQL Server 2025
    ↓
Enterprise Developer
```

TEXT (CONT.)

```
    ↓
SSMS
    ↓
Practice databases
```

Remember:

Developer Edition is free for development/testing, but it must not be used as the production database server.

SQL Server Express Edition

SQL Server Express is a free, smaller SQL Server edition.

It is intended for:

TEXT

```
Learning
Small applications
Small desktop applications
Small server applications
Developer projects
```

Microsoft describes Express as its free entry-level SQL Server edition.

SQL Server 2025 Express Limits

SQL Server 2025 Express has important resource limits.

Its Database Engine is limited to the lesser of:

TEXT

```
1 socket
or
4 cores
```

Its buffer pool memory limit is approximately:

TEXT

```
1,410 MB
```

and SQL Server 2025 increased the maximum relational database size to:

TEXT

```
50 GB
```

per database.

Older SQL Server Express versions had a smaller database-size limit, so this is an important SQL Server

2025 change.

Developer Edition vs Express

Developer Edition	Express Edition
Free for development/test	Free
Large SQL Server feature set	Limited feature/resource set
Great for learning advanced SQL Server	Good for small applications
Not licensed for production	Can be used for appropriate small production workloads
Enterprise Developer mirrors Enterprise features	Resource limits apply
SQL Server Agent available	SQL Server Agent unavailable

SQL Server 2025 Express does not include SQL Server Agent.

For your learning:

```text
Want to learn full SQL Server?
    ↓
Developer Edition

Want a lightweight free database
for a small application?
    ↓
Express
```

SQL Server Management Studio — SSMS

SSMS stands for:

> **SQL Server Management Studio**

SSMS is a graphical client and management tool used to connect to and work with SQL Server.

Simple definition:

> **SSMS is the application you use to communicate with and manage SQL Server.**

It is not:

```text
A database
A SQL Server instance
The Database Engine
```

It is a client application.

```text
SSMS
↓
SQL Server Instance
↓
Database Engine
↓
Database
```

The current general-availability generation is SSMS 22; as of September 2026, Microsoft's current release is SSMS 22.9.2.

What Can We Do Using SSMS?

SSMS allows you to:

```text
Connect to SQL Server

Create databases

Create tables

Write SQL

Execute queries

View query results

Manage users and permissions

View SQL Server objects

Create SQL Server Agent jobs

Monitor server activity

Perform backups and restores

Perform administrative work
```

For example, in SSMS:

```sql
SELECT *
FROM Employees;
```

SSMS sends the query to SQL Server:

```text
SSMS
↓
```

TEXT (CONT.)

```
SQL query
↓
Database Engine
↓
Result
↓
SSMS
```

Connecting to SQL Server

Before working with SQL Server through SSMS, you must connect to a SQL Server instance.

The connection window asks for information such as:

TEXT

```
Server name
Authentication
User name
Password
Encryption options
```

Microsoft's current SSMS connection experience supports Windows Authentication, SQL Server Authentication, and supported Microsoft Entra authentication methods.

For your SQL Server fundamentals, focus first on:

TEXT

```
Windows Authentication

SQL Server Authentication
```

Server Name

The Server name tells SSMS:

Which SQL Server instance do you want to connect to?

Suppose your computer name is:

TEXT

```
DEV-PC
```

and SQL Server is installed as the default instance.

You might connect using:

```
DEV-PC
```

or locally:

```
localhost
```

For a named instance:

```
DEV-PC\SQLDEV
```

Here:

```
DEV-PC
=
Computer/server name

SQLDEV
=
SQL Server instance name
```

Microsoft documents connection names such as:

```
MyServer

MyServer\MyInstance

localhost
```

depending on the installation.

Default Instance Server Name

Suppose your machine is:

```
DEV-PC
```

and SQL Server has a default instance.

You normally connect with:

```
DEV-PC
```

or locally:

```text
localhost
```

Conceptually:

```text
DEV-PC
 ↓
Default SQL Server instance
```

Named Instance Server Name

Suppose the instance is called:

```text
SQLDEV
```

Then:

```text
DEV-PC\SQLDEV
```

means:

```text
Computer
 \
Instance
```

Think:

```text
ServerName\InstanceName
```

Authentication

After finding the correct SQL Server instance, SQL Server must determine:

> **Who are you?**

That process is called:

> **Authentication**

Simple definition:

Authentication verifies the identity of the person or application connecting to SQL Server.

For example:

```text
You connect
  ↓
SQL Server asks:
"Who are you?"
  ↓
Authentication
```

SQL Server has two server authentication modes:

```text
Windows Authentication Mode

Mixed Mode
(Windows + SQL Server Authentication)
```

Windows Authentication is always available; Mixed Mode additionally enables SQL Server Authentication.

Windows Authentication

With Windows Authentication, SQL Server uses your Windows identity.

Suppose you logged into Windows as:

```text
COMPANY\Jayanth
```

When you select:

```text
Windows Authentication
```

SSMS can use that existing Windows identity.

Conceptually:

```text
Login to Windows
   ↓
Windows knows your identity
   ↓
SSMS connects
   ↓
SQL Server trusts the Windows identity
```

You normally don't type another SQL Server username/password into SSMS.

Microsoft recommends Windows Authentication when possible.

Why Windows Authentication Is Preferred

Windows handles things such as:

```text
Password policies
Account lockout
Domain identities
Group membership
Credential management
Kerberos/NTLM authentication
```

This avoids maintaining another password separately inside SQL Server.

For corporate environments:

```text
Company Windows account
    ↓
SQL Server access
```

is very common.

SQL Server Authentication

With SQL Server Authentication, the username and password are maintained by SQL Server instead of using your Windows account.

For example:

```text
Login:
app_user

Password:
```

Conceptually:

```text
SSMS
↓
Username + Password
↓
SQL Server
↓
Validate SQL login
```

SQL Server Authentication requires the server to be configured for:

SQL Server and Windows Authentication Mode, commonly called Mixed Mode.

Example SQL Server Login

A privileged administrator could create a SQL login:

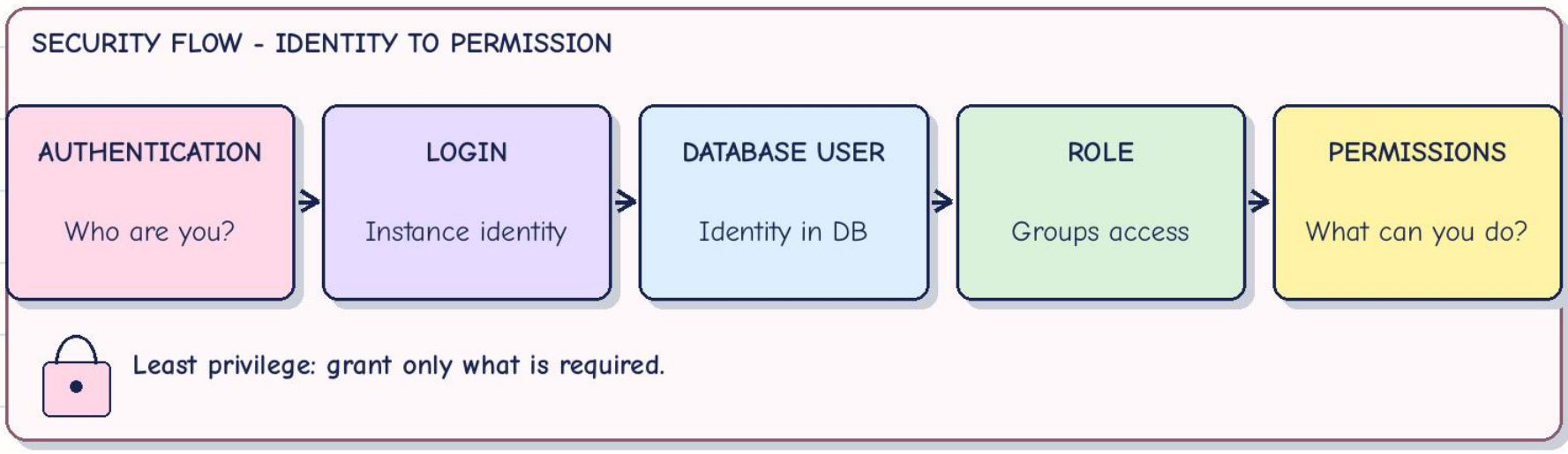
```sql
CREATE LOGIN AppLogin
WITH PASSWORD = 'Use-A-Strong-Password-Here';
```

The login allows authentication at the SQL Server instance level.

But this does not automatically mean that the login can access every database.

That leads to an important security concept.

Authentication vs Authorization



These are different.

Authentication

Authentication asks:

Who are you?

Example:

```text
Windows Authentication
SQL Server Authentication
```

Authorization

Authorization asks:

What are you allowed to do?

For example:

```text
Can you read Employees?

Can you modify Orders?

Can you delete Customers?

Can you create tables?
```

Easy memory:

```text
Authentication
=
Who are you?

Authorization
=
What can you do?
```

Windows Authentication vs SQL Server Authentication

Windows Authentication	SQL Server Authentication
Uses Windows identity	Uses SQL Server login
Usually no separate SQL password entered	Requires SQL username/password
Recommended where possible	Useful when Windows identity isn't suitable
Integrated with Windows security	Credentials managed by SQL Server
Works in Windows/domain environments	Can support applications/users outside Windows trust scenarios

Microsoft recommends Windows Authentication where possible because it provides stronger integration with Windows account security.

What Is Mixed Mode?

Mixed Mode means SQL Server permits both:

```text
Windows Authentication
+
SQL Server Authentication
```

So:

```text
Windows-only mode
→ Windows logins

Mixed mode
→ Windows logins + SQL Server logins
```

Changing between these modes requires SQL Server configuration, and switching authentication modes requires restarting the SQL Server service.

What About the sa Login?

sa means the built-in SQL Server system administrator login.

It uses SQL Server Authentication.

Because sa is well known and commonly targeted, Microsoft recommends not enabling or using it unless necessary.

For applications, don't think:

```text
Application should always connect using sa.
```

That is poor practice.

Applications should generally receive only the permissions they require.

Object Explorer

Object Explorer is the tree-like navigation area in SSMS.

After connecting, you may see:

```text
DEV-PC\SQLDEV
│
├── Databases
├── Security
├── Server Objects
├── Replication
├── Management
└── SQL Server Agent
```

Object Explorer lets you visually browse and manage SQL Server objects.

Databases in Object Explorer

Expand:

```text
Databases
```

and you might see:

```text
Databases
│
├── System Databases
├── EmployeeDb
├── ShoppingDb
└── FundraisingDb
```

Expand:

```text
EmployeeDb
```

and you may see:

```text
EmployeeDb
│
├── Tables
├── Views
├── Programmability
├── Security
└── Storage
```

Object Explorer is therefore essentially a graphical way to navigate your SQL Server environment.

Object Explorer vs Query Editor

They have different purposes.

Object Explorer

Used to:

```text
Browse objects
View databases
View tables
Manage security
Open properties
Manage jobs
```

Query Editor

Used to write SQL:

```sql
SELECT *
FROM Employees;
```

Think:

```text
Object Explorer
=
Navigate visually

Query Editor
=
Write SQL
```

Database Security

SQL Server security can initially seem confusing because several concepts work together.

For basic learning, understand this flow:

```text
Login
↓
User
↓
Role / Permissions
↓
Database Objects
```

Login

A login generally represents an identity that can authenticate to the SQL Server instance.

Examples:

```text
Windows account
Windows group
SQL Server login
```

Think:

```text
Login
=
Can I connect to this SQL Server instance?
```

Microsoft notes that server logins and database users are separate concepts.



Database User

A database user represents an identity inside a particular database.

Suppose:

```text
AppLogin
```

can connect to SQL Server.

You can map it to a user in:

```text
EmployeeDb
```

For example:

```sql
USE EmployeeDb;
GO

CREATE USER AppUser
FOR LOGIN AppLogin;
GO
```

Now:

```text
AppLogin
  ↓
SQL Server login
  ↓
AppUser
  ↓
EmployeeDb user
```

Permissions

Permissions determine what the user can do.

Examples:

```text
SELECT
INSERT
UPDATE
DELETE
EXECUTE
```

For example:

```sql
GRANT SELECT
ON dbo.Employees
TO AppUser;
```

Now AppUser can read Employees.

Think:

```text
Login
=
Enter SQL Server

User
=
Identity inside database

Permission
=
What that identity can do
```

Database Roles

Instead of giving every user permissions individually, users can be placed into roles.

Conceptually:

```text
Users
↓
Role
↓
Permissions
```

For example:

```sql
ALTER ROLE db_datareader
ADD MEMBER AppUser;
```

This adds AppUser to the built-in database role that provides read access to user tables/views in the database.

For real applications, follow the least privilege principle:

> **Give users and applications only the permissions they actually require.**

Microsoft's current security guidance recommends mapping logins to users/roles and granting only required access.

SQL Server Security in Object Explorer

At the server level:

```text
SQL Server
│
└── Security
   │
   ├── Logins
   ├── Server Roles
   └── Credentials
```

Inside an individual database:

```text
EmployeeDb
│
└── Security
   │
   ├── Users
   ├── Roles
   └── Schemas
```

The difference matters:

```text
Server Security
=
SQL Server instance level

Database Security
=
Individual database level
```

SQL Server Agent

SQL Server Agent is a service used to automate and schedule SQL Server administrative tasks.

Simple definition:

> **SQL Server Agent runs automated tasks, called jobs.**

Microsoft describes SQL Server Agent as a Windows service that executes scheduled administrative tasks called jobs.

For example:

```text
Every day at 2:00 AM
    ↓
```

TEXT (CONT.)

```
Backup database
```

Instead of a person manually running the backup every night:

TEXT

```
SQL Server Agent
    ↓
Automatically runs job
```

SQL Server Agent Job

A job is a task or collection of steps that SQL Server Agent executes.

Examples:

TEXT

```
Database backup

Run stored procedure

Import data

Delete old records

Perform maintenance

Generate scheduled processing
```

A job can be run:

TEXT

```
Manually

On a schedule

In response to certain events
```

Job Structure

Think:

TEXT

```
SQL Server Agent
   ↓
   Job
   ↓
 Job Steps
   ↓
```

TEXT (CONT.)

```
  Schedule
```

For example:

TEXT

```
Job:
NightlyDatabaseBackup

Step:
Run backup command

Schedule:
Every day at 2:00 AM
```

Job Steps

A job contains at least one step.

A job could contain:

TEXT

```
Step 1
Backup EmployeeDb

Step 2
Backup CustomerDb

Step 3
Perform cleanup
```

SQL Server Agent executes the steps according to the job configuration.

Microsoft's SQL Server Agent documentation notes that jobs can contain multiple steps and schedules.

Job Schedule

The schedule determines when a job runs.

Examples:

TEXT

```
Every day at 2 AM

Every Monday

Every hour

When SQL Server Agent starts
```

Example:

```
Job
NightlyBackup
   ↓
Schedule
Daily at 2:00 AM
```

Job History

SQL Server Agent records information about job execution.

You can check:

```
Did the job run?

Did it succeed?

Did it fail?

When did it run?

Which step failed?
```

This information is very useful when troubleshooting automated processes.

SQL Server Agent in Object Explorer

In SSMS:

```
Server
│
└── SQL Server Agent
   │
   ├── Jobs
   ├── Job Activity Monitor
   ├── Alerts
   ├── Operators
   └── Proxies
```

You don't need to learn every item immediately.

For now:

```
Jobs
=
Automated tasks

Job Activity Monitor
```

TEXT (CONT.)

```
=
Monitor those jobs
```

Important: Express Does Not Include SQL Server Agent

If you install SQL Server Express, you may wonder:

> **Why don't I see SQL Server Agent?**

Because SQL Server 2025 Express does not include SQL Server Agent. Standard and Enterprise do.

Developer editions inherit the corresponding Enterprise/Standard features for development and testing, so SQL Server Agent is available there.

This is another reason Developer Edition is convenient for learning SQL Server.

Activity Monitor

Activity Monitor is an SSMS feature used to see what is currently happening inside a SQL Server instance.

Simple definition:

> **Activity Monitor provides a graphical view of current SQL Server processes, workload, and resource activity.**

You can open it from SSMS for a connected SQL Server instance.

Microsoft's current Activity Monitor contains areas including:

TEXT

```
Overview

Processes

Resource Waits

Data File I/O

Recent Expensive Queries

Active Expensive Queries
```

Overview

The Overview section provides a quick view of current activity.

It can help you notice things such as:

```text
Processor activity

Waiting tasks

Database I/O

Batch activity
```

Think:

```text
Overview
=
Quick health/activity view
```

Processes

The Processes section shows active sessions/processes connected to SQL Server.

It can help identify:

```text
Who is connected?

What session is running?

Is a session blocked?

What database is being used?
```

This is useful when investigating blocking or active workloads.

Resource Waits

SQL Server sometimes has work that must wait for a resource.

Examples could involve:

```text
CPU
Locks
Disk I/O
Memory-related resources
```

The Resource Waits section gives a graphical view of current waits.

It helps answer:

> **What resources are SQL Server requests currently waiting for?**

Data File I/O

The Data File I/O section helps show activity associated with database files.

Think:

```text
Database files
   ↓
Read/write activity
```

This can help during performance troubleshooting.

Recent Expensive Queries

This area shows queries that have recently consumed significant resources.

For example, you might discover a query using a large amount of:

```text
CPU
I/O
Execution resources
```

It can be useful for quick performance investigation.

Active Expensive Queries

This shows expensive queries that are currently active.

Difference:

```text
Recent Expensive Queries
=
Recently executed

Active Expensive Queries
=
Currently executing
```

Activity Monitor vs Job Activity Monitor

Do not confuse these.

Activity Monitor

Used for:

```
SQL Server processes
Performance
Resource waits
I/O
Expensive queries
Blocking
```

Job Activity Monitor

Used for:

```
SQL Server Agent jobs

Which job is running?

Which job succeeded?

Which job failed?

When did a job run?
```

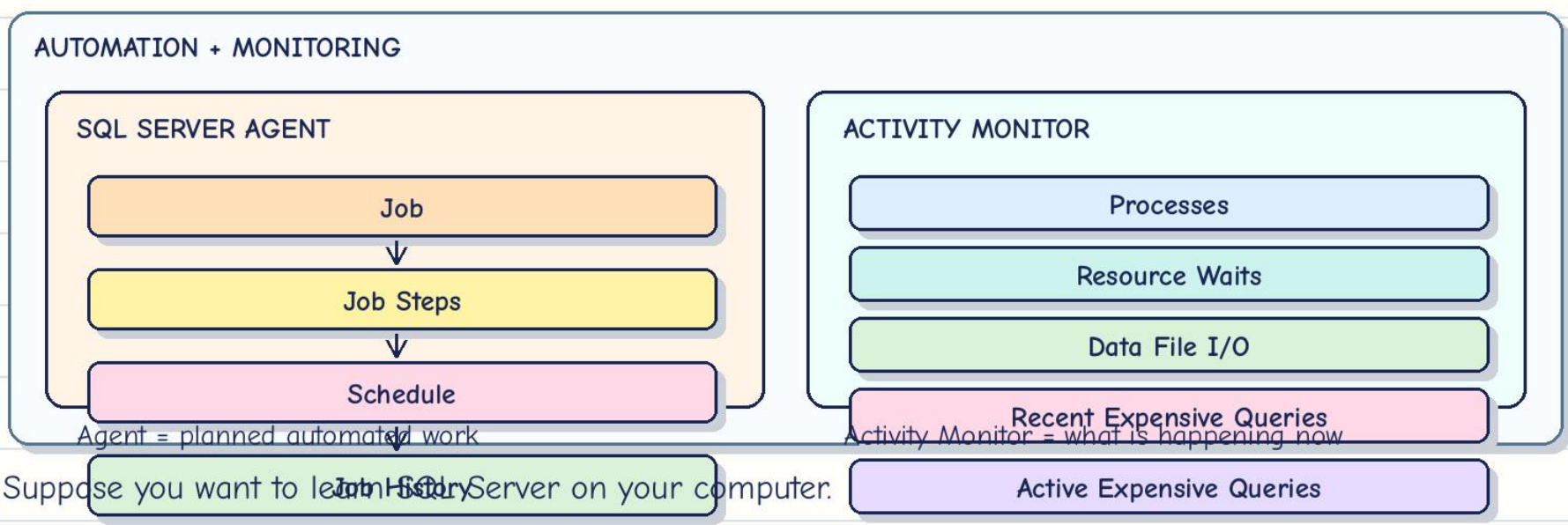
Microsoft's Job Activity Monitor supports viewing current job state, history, starting/stopping jobs, and related job information.

Easy memory:

```
Activity Monitor
=
Server workload

Job Activity Monitor
=
Agent jobs
```

Putting Everything Together



Suppose you want to learn SQL Server on your computer.

You install:

```text
SQL Server 2025
Enterprise Developer Edition
```

Then install:

```text
SSMS 22
```

Suppose your SQL Server instance is:

```text
DEV-PC\SQLDEV
```

You open SSMS.

You enter:

```text
Server Name:
DEV-PC\SQLDEV

Authentication:
Windows Authentication
```

Then:

```text
SSMS
↓
Authenticates your Windows identity
↓
Connects to DEV-PC\SQLDEV
↓
Object Explorer appears
```

You might see:

```text
DEV-PC\SQLDEV
│
├── Databases
│   └── EmployeeDb
│
├── Security
│   └── Logins
│
├── Management
│
└── SQL Server Agent
  └── Jobs
```

You execute:

```sql
USE EmployeeDb;
GO

SELECT *
FROM Employees;
```

The Database Engine processes the query and SSMS displays the result.

If you want to schedule something:

```text
SQL Server Agent
↓
Create Job
↓
Create Job Step
↓
Create Schedule
```

If performance suddenly becomes slow:

```text
Activity Monitor
↓
Processes
Resource Waits
Expensive Queries
```

That is how all these concepts connect.

Final Comparison

Concept	Meaning
SQL Server Version	Generation, such as SQL Server 2025
SQL Server Edition	Feature/licensing package
Enterprise	Highest-end production edition
Standard	General production edition
Developer	Development/testing edition
Express	Free lightweight edition with limits
SSMS	Client/management application
Server Name	Identifies the SQL Server instance to connect to
Authentication	Verifies who you are

Concept	Meaning
Windows Authentication	Uses Windows identity
SQL Server Authentication	Uses SQL Server username/password
Object Explorer	Graphical SQL Server object navigator
Login	Identity that can connect to SQL Server
Database User	Identity inside a database
Permission	Defines allowed actions
SQL Server Agent	Automation/scheduling service
Job	Automated task
Activity Monitor	Current SQL Server workload/performance viewer

Easy Memory Rule

```text
Edition
=
Which SQL Server package am I using?
```

```text
SSMS
=
Tool I use to connect
```

```text
Server Name
=
Where am I connecting?
```

```text
Authentication
=
Who am I?
```

```text
Security
=
What am I allowed to do?
```

```text
SQL Server Agent
=
What should SQL Server run automatically?
```

```text
Activity Monitor
=
What is SQL Server doing right now?
```

And the complete flow:

```text
SQL Server Edition
    ↓
SQL Server Instance
    ↑
   SSMS
    ↓
Server Name
    ↓
Authentication
    ↓
Object Explorer
    ↓
Database + Security
    ↓
SQL Server Agent / Activity Monitor
```

Interview Questions

1. What is a SQL Server edition?

A SQL Server edition is a particular feature and licensing package of SQL Server.

For example:

```text
Enterprise
Standard
Developer
Express
```

Interviewer is testing: Whether you understand version versus edition.

2. What is the difference between SQL Server version and edition?

A version represents the generation, such as SQL Server 2025.

An edition represents the package, such as Enterprise or Express.

```text
SQL Server 2025 Express

2025 = Version
Express = Edition
```

3. What is SQL Server Developer Edition?

Developer Edition provides SQL Server capabilities for development and testing without being licensed for production deployment.

SQL Server 2025 offers Enterprise Developer and Standard Developer editions.

4. Can Developer Edition be used in production?

No.

It is licensed for:

```text
Development
Testing
```

not production.

5. Why is Developer Edition useful for learning?

It lets developers practice substantial SQL Server functionality without purchasing a production Enterprise license.

Enterprise Developer is therefore a strong option for SQL Server learning.

6. What is SQL Server Express?

SQL Server Express is Microsoft's free entry-level SQL Server edition designed for learning and smaller applications.

It has resource and feature limits compared with Standard and Enterprise.

7. What is the maximum database size in SQL Server 2025 Express?

The maximum relational database size is:

```text
50 GB
```

per database.

8. Does SQL Server Express support SQL Server Agent?

No.

SQL Server Agent is not included with Express Edition.

9. What is SSMS?

SSMS stands for SQL Server Management Studio.

It is a graphical client and management environment used to connect to SQL Server, execute SQL, browse database objects, and perform administrative work.

10. Is SSMS the SQL Server Database Engine?

No.

```text
SSMS
=
Client/tool

Database Engine
=
Server component processing database work
```

11. What information is normally needed to connect from SSMS?

Common connection information includes:

```text
Server name
Authentication method
Credentials when required
```

Modern SSMS also provides encryption-related connection options.

12. What does Server Name mean?

Server Name tells SSMS which SQL Server instance it should connect to.

Example:

```text
SERVER01
```

or:

```text
SERVER01\SQLDEV
```

13. What does SERVER01\SQLDEV mean?

```text
SERVER01
=
Computer/server

SQLDEV
=
Named SQL Server instance
```

Together they identify a specific SQL Server instance.

14. What is authentication?

Authentication is the process of verifying the identity of a user or application attempting to connect.

Think:

```text
Authentication
=
Who are you?
```

15. What is Windows Authentication?

Windows Authentication allows SQL Server to use an authenticated Windows identity.

It integrates SQL Server access with Windows/domain security and is recommended where possible.

16. What is SQL Server Authentication?

SQL Server Authentication uses a login name and password maintained by SQL Server.

Example:

```text
Login: ApplicationUser
Password:
```

It requires SQL Server to allow SQL authentication through Mixed Mode.

17. What is Mixed Mode Authentication?

Mixed Mode allows both:

```text
Windows Authentication
+
SQL Server Authentication
```

18. What is the difference between authentication and authorization?

Authentication verifies:

```text
Who are you?
```

Authorization determines:

```text
What are you allowed to do?
```

This is one of the most important security distinctions.

19. What is Object Explorer?

Object Explorer is the tree-based interface in SSMS used to browse SQL Server objects.

For example:

```text
Server
↓
Databases
↓
Database
↓
Tables
```

20. What is the difference between a login and a database user?

A login normally represents an identity at the SQL Server instance level.

A database user represents an identity inside a specific database.

```text
Login
↓
SQL Server Instance

User
↓
Database
```

21. What are permissions?

Permissions define what a user or login is allowed to do.

Examples include:

```text
SELECT
INSERT
UPDATE
DELETE
```



TEXT (CONT.)

```
EXECUTE
```

22. What is SQL Server Agent?

SQL Server Agent is the service responsible for executing automated and scheduled SQL Server tasks called jobs.

Example:

TEXT

```
Every night at 2 AM
    ↓
Database backup job
```

23. What are a job, job step, and schedule?

A job is the overall automated task.

A job step is an individual action inside the job.

A schedule determines when the job runs.

TEXT

```
Job
↓
Steps
↓
Schedule
```

24. What is Activity Monitor?

Activity Monitor is an SSMS tool that provides a graphical view of current SQL Server workload and performance information.

It includes areas for:

TEXT

```
Processes
Resource Waits
Data File I/O
Recent Expensive Queries
Active Expensive Queries
```

25. What is the difference between Activity Monitor and Job Activity Monitor?

Activity Monitor monitors SQL Server workload and performance.

Job Activity Monitor monitors SQL Server Agent jobs.

```text
Activity Monitor
=
What is SQL Server doing?

Job Activity Monitor
=
What are my Agent jobs doing?
```

That distinction is frequently useful during SQL Server administration.

Quick Revision

```text
SQL Server 2025
=
Version
```

```text
Enterprise / Standard / Developer / Express
=
Editions
```

```text
Developer
=
Development + testing

Not production
```

```text
Express
=
Free lightweight edition
+
Resource limits
+
No SQL Server Agent
```

```text
SSMS
=
SQL Server client/management tool
```

TEXT
Server Name
=
Which SQL Server instance?

TEXT
Authentication
=
Who are you?

TEXT
Windows Authentication
=
Use Windows identity

TEXT
SQL Server Authentication
=
Use SQL Server login + password

TEXT
Authorization
=
What can you do?

TEXT
Object Explorer
=
Browse/manage SQL Server objects

TEXT
SQL Server Agent
=
Automate jobs

TEXT
Activity Monitor
=
View current SQL Server activity/performance

Final Memory Diagram

FINAL MEMORY DIAGRAM

